Group II, Row IV (final-state $g \to gg$): from Eq. (0.4) of my note to the final result of the collaborator's note

1 Conventions and dictionary

- "(0.n)" = equations of `Group_II_Row_IV.tex`; "N(n)" = equations of the collaborator's note (23-page version, with Sec. 9 "All subtractions in coordinate space").
- ϵ below is the note's: $d = 2 - 2\epsilon$. In my note $d = 2 - \epsilon_{\rm me}$, so $\epsilon_{\rm me} = 2\epsilon$ and $\frac{1}{\epsilon} = \frac{2}{\epsilon_{\rm me}}$. Polarisation sum $d_\perp = d$ (my $d_\perp = 2 - \epsilon_{\rm me}$ is the same thing). Collinear poles are IR, $\epsilon < 0$.
- Prefactor, N(8): $\bar{C} = \dfrac{1}{(2\pi)^3}\dfrac{g^4 N_c}{4\pi^4}\dfrac{1}{2\pi} = \dfrac{1}{(2\pi)^3}\dfrac{g^4 N_c}{8\pi^5}$, $\qquad \dfrac{\pi\bar{C}}{k^+} = \dfrac{1}{(2\pi)^3}\dfrac{g^4 N_c}{8\pi^4}\dfrac{1}{k^+} = \dfrac{1}{(2\pi)^3}\dfrac{g^2}{2\pi^2}\dfrac{g^2}{8\pi^2}\dfrac{2N_c}{k^+}$.
- LO amplitude, N(4), and the product that appears everywhere:

$$A_i^c(y) \equiv \int_x \frac{(x-y)_i}{(x-y)^2}\big[U(y)-U(x)\big]^{ce}\rho^e(x),$$

$$A_k^c(y)A_{k'}^c(y') = \int_{x,x'} \frac{(x-y)_k(x'-y')_{k'}}{(x-y)^2(x'-y')^2}\,\big[U^{ce'}(y')-U^{ce'}(x')\big]\big[U^{ce}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x). \tag{1.1}$$

  In particular $A(y)\cdot A(y') = \int_{x,x'} \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y')-U^{ce'}(x')\big]\big[U^{ce}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x)$.
- $\hat{P}(\xi) = \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi)$, $\quad c(\xi) = \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}$, $\quad \int_0^1 d\xi\, \frac{\xi}{(1-\xi)_+}h(\xi) \equiv \int_0^1 d\xi\, \frac{\xi h(\xi)-h(1)}{1-\xi}$, $\hat{P}_+ = \hat{P}$ with this plus.

2 What is wrong in my file

(A) The main problem: the wrong variable is continued to d dimensions [(0.14)→(0.15)]. In (0.14) the δ-functions are used to integrate out w, w'. This is done in $d = 2$: Jacobian ξ^{-2} per $\delta^{(2)}$, and the kernels rescale as $\frac{(w-z)^m}{(w-z)^2} = \xi\frac{(y-z)^m}{(y-z)^2}$. Then in (0.15) the remaining z integral is continued to d dimensions *at fixed* y, y':

$$\int_z \frac{(y-z)^m(y'-z)^{m'}}{(y-z)^2(y'-z)^2} \;=\; H^{mm'}(y'-y) \;=\; -\frac{\pi}{2}\Big[\frac{2}{\epsilon_{\rm me}} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}\Big]\delta^{mm'} - \pi\,\hat{s}^m\hat{s}^{m'}. \tag{2.1}$$

The note continues instead the *parent-to-daughter relative coordinate* $r = w - z$ (equivalently the relative momentum ℓ of the pair, N(18), Sec. 9.2), at fixed $u = w' - w = (y'-y)/\xi$. That gives $H^{mm'}(u)$, not $H^{mm'}(y'-y) = H^{mm'}(\xi u)$. In $d = 2$ the two are the same (formally divergent) integral, related by $z = y - \xi r$. In dimensional regularisation they are not: the exact form N(71) gives

$$\boxed{H^{mm'}(\xi u) = \xi^{2\epsilon}\,H^{mm'}(u)} \qquad\Longrightarrow\qquad \frac{\xi^{2\epsilon}}{\epsilon} = \frac{1}{\epsilon} + \log\xi^2 + O(\epsilon). \tag{2.2}$$

So in every $\xi \neq 1$ term my collinear bracket must be

$$\Big[\frac{2}{\epsilon_{\rm me}} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}\Big] \;\longrightarrow\; \Big[\frac{2}{\epsilon_{\rm me}} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4\xi^2}\Big]. \tag{2.3}$$

The poles are unchanged; the finite part of my result differs from the note's by

$$\begin{aligned}\frac{d^3N}{d^3k}\Big|_{\rm mine} - \frac{d^3N}{d^3k}\Big|_{\rm note} =& -\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\,\hat{P}(\xi)\,\frac{\log\xi^2}{\xi^2}\\ &\times\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y')-U^{ce'}(x')\big]\big[U^{ce}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x).\end{aligned} \tag{2.4}$$

It vanishes at $\xi = 1$ (so JIMWLK, the $\delta(1-\xi)$ term and the running coupling are unaffected) and at $\xi \to 0$ (so BFKL is unaffected). Which variable is continued is part of the definition of the $\overline{\rm MS}$ scheme; the note's counterterms (its Sec. 7) are built on the r (or ℓ) continuation, so to match the note one must integrate out y, y' (not w, w') with the δ's and continue r. This is done in Sec. 3.

(B) Bookkeeping differences (not errors, but they must be changed to match the note).

1. (0.5), (0.8), (0.9): I subtract the endpoints with weights $\frac{\xi}{1-\xi}$ and $\frac{1-\xi}{\xi}$. The note subtracts with $\frac{1}{1-\xi}$ and $\frac{1}{\xi}$, N(24). Since $\frac{\xi}{1-\xi}=\frac{1}{1-\xi}-1$ and $\frac{1-\xi}{\xi}=\frac{1}{\xi}-1$, my choice produces the extra boundary terms $\propto[\frac{1}{\vee}-\frac{1}{k^+ +\Lambda}]$ in (0.5). They cancel identically against $\int d\xi\,(\mathcal{J}+\mathcal{B})$ in my remainder.
2. I keep $O(k^+/\vee)$ terms (lower limits $k^+/\vee$, $\frac{1}{\vee}$, $\log(1-\frac{k^+}{\vee})$); the note drops them, N(27).
3. My DGLAP and RC terms carry finite parts. In the note the DGLAP and RC entries are the poles only; all finite parts are in $F(k;\mu)$, N(34). The finite parts of $\frac{\beta_0}{2}\delta(1-\xi)$ and of my RC term are equal and opposite, so moving them is harmless.
4. The block called "Finite" in (0.12)–(0.15) contains $\frac{2}{\epsilon}$ poles. In (0.19) the "finite" term contains $-(1-\frac{k^+}{\vee})\mathcal{B}$, which is $O(1)$, not a power correction, and carries the net-colour pole.
5. The net-colour IR pole (N Sec. 8, 9.7) is not separated.

(C) Minor. In (0.7) the antisymmetric structure $2(\delta_{mk'}\delta_{km'}-\delta_{km}\delta_{k'm'})$ is dropped before the z integral; it vanishes only after it (because $H^{mm'}$ is symmetric), so say so. Typos: (0.2), (0.3) have commas instead of \, in $\frac{1}{(2\pi)^3}$, $\frac{g^4}{4\pi^4}$, and $\frac{1}{1-\xi},\delta_{jm}$; "JIMLWK".

3 Derivation, step by step

Step 1. Starting point: Eq. (0.4) = N(1), N(2)

$$\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{4\pi^4}\,\frac{1}{k^+}\int_{\xi_\vee}^{\xi_\Lambda}\frac{d\xi}{2\pi}\int_{z,w,w',x'}e^{-ik\cdot(w'-w)}\int_{x,y,y'}\frac{(w-z)^m(x-y)^k(w'-z)^{m'}(x'-y')^{k'}}{(w-z)^2(x-y)^2(w'-z)^2(x'-y')^2}$$
$$\times\,\delta^{(2)}\big[y-z-\xi(w-z)\big]\,\delta^{(2)}\big[y'-z-\xi(w'-z)\big]\,T_{mkm'k'}(\xi)\,\big[U^{ce'}(y')-U^{ce'}(x')\big]\big[U^{ce}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x),\tag{3.1}$$

$$T_{mkm'k'}(\xi)=d_\perp\,\xi(1-\xi)\,\delta_{km}\delta_{k'm'}+2\big(\delta_{mk'}\delta_{km'}-\delta_{km}\delta_{k'm'}\big)+c(\xi)\,\delta_{mm'}\delta_{kk'},\qquad \xi_\vee=\frac{k^+}{\vee},\;\;\xi_\Lambda=\frac{k^+}{k^++\Lambda}.$$

This is exactly my (0.4) (the upper limit $k^+/(k^++\Lambda)$, the colour structure and T are identical).

Step 2. Integrate x,x': LO amplitudes

By (1.1) the x,x' integrals and the colour structure are $A^c_k(y)A^c_{k'}(y')$:

$$\frac{d^3N}{d^3k}=\frac{\bar C}{k^+}\int_{\xi_\vee}^{\xi_\Lambda}d\xi\int_{z,w,w',y,y'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)^m(w'-z)^{m'}}{(w-z)^2(w'-z)^2}\,\delta^{(2)}\big[y-z-\xi(w-z)\big]\delta^{(2)}\big[y'-z-\xi(w'-z)\big]\,T_{mkm'k'}\,A^c_k(y)A^c_{k'}(y').\tag{3.2}$$

Step 3. Integrate y,y' with the δ's (not w,w')

$\int_y\delta^{(2)}[y-z-\xi(w-z)]\,g(y)=g(z+\xi(w-z))$, Jacobian 1, and the splitting kernel $(w-z)^m/(w-z)^2$ is *not* rescaled. Result, N(7):

$$\frac{d^3N}{d^3k}=\frac{\bar C}{k^+}\int_{\xi_\vee}^{\xi_\Lambda}d\xi\;I(\xi),\qquad I(\xi)=\int_{z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)^m(w'-z)^{m'}}{(w-z)^2(w'-z)^2}\,T_{mkm'k'}\,A^c_k(y)A^c_{k'}(y')\Big|_{y=z+\xi(w-z),\;y'=z+\xi(w'-z)}.\tag{3.3}$$

Step 4. Variables (r,u,Y), N(64)–N(67)

Define $r=w-z$, $u=w'-w$, $Y=\frac12(y+y')$. Then $w'-z=r+u$, $y'-y=\xi u$ and

$$y=Y-\tfrac{\xi}{2}u,\qquad y'=Y+\tfrac{\xi}{2}u,\qquad z=Y-\xi\big(r+\tfrac12u\big),\qquad w=z+r,\qquad w'=w+u.\tag{3.4}$$

Per transverse component $\partial(z,w,w')/\partial(Y,r,u) = \begin{pmatrix} 1 & -\xi & -\xi/2 \\ 1 & 1-\xi & -\xi/2 \\ 1 & 1-\xi & 1-\xi/2 \end{pmatrix}$, whose determinant is 1, so $d^2z\,d^2w\,d^2w' = d^2Y\,d^2u\,d^2r$. The phase is $e^{-ik\cdot u}$ and Y sits only in the LO amplitudes:

$$\boxed{I(\xi) = \int d^2u\; e^{-ik\cdot u}\; T_{mkm'k'}(\xi)\; H^{mm'}(u)\; C_{kk'}(\xi u)} \tag{3.5}$$

$$H^{mm'}(u) \equiv \int_r \frac{r^m}{r^2}\frac{(r+u)^{m'}}{(r+u)^2}, \tag{3.6}$$

$$\begin{aligned} C_{kk'}(v) &\equiv \int_Y A^c_k\big(Y-\tfrac{v}{2}\big)A^c_{k'}\big(Y+\tfrac{v}{2}\big) \\ &= \int_{Y,x,x'} \frac{(x-Y+\frac{v}{2})_k\,(x'-Y-\frac{v}{2})_{k'}}{(x-Y+\frac{v}{2})^2(x'-Y-\frac{v}{2})^2}\,\big[U^{ce'}(Y+\frac{v}{2})-U^{ce'}(x')\big]\big[U^{ce}(Y-\frac{v}{2})-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \end{aligned} \tag{3.7}$$

Back to (y,y'). Since $y = Y-\frac{\xi}{2}u$, $y' = Y+\frac{\xi}{2}u$ give $d^2y\,d^2y' = \xi^2 d^2Y\,d^2u$, for any F:

$$\int d^2u\,e^{-ik\cdot u}F(u)\,C_{kk'}(\xi u) = \frac{1}{\xi^2}\int_{y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,F\Big(\frac{y'-y}{\xi}\Big)A^c_k(y)A^c_{k'}(y'). \tag{3.8}$$

Step 5. Dimensional regularisation: continue only r, N Sec. 9.2

$r\in\mathbb{R}^d$, $d=2-2\epsilon$; u, Y and the 2D fields stay two-dimensional. The prescription that reproduces the momentum-space $\overline{\rm MS}$ of N(18) is

$$\frac{r^m}{r^2}\to\frac{r^m}{|r|^d},\qquad d^2r\to\mathcal{N}_\epsilon\,d^dr,\qquad \mathcal{N}_\epsilon=\Gamma(1-\epsilon)^2\Big(\frac{\pi\mu^2e^{\gamma_E}}{4}\Big)^\epsilon,\qquad H^{mm'}(u)=\mathcal{N}_\epsilon\int d^dr\,\frac{r^m}{|r|^d}\frac{(r+u)^{m'}}{|r+u|^d}. \tag{3.9}$$

The integrand falls like $|r|^{-d}$: the large-$|r|$ (collinear) region converges for $\epsilon<0$; $r\to0$ and $r\to-u$ are integrable.

Step 6. $H^{mm'}(u)$ exactly, N(68)–N(72)

Write $\frac{r^m}{|r|^d}=\partial_m\phi(r)$ with $\phi(r)=\frac{|r|^{2\epsilon}}{2\epsilon}$, $\nabla^2\phi = S_d\,\delta^{(d)}(r)$, $S_d=\frac{2\pi^{d/2}}{\Gamma(d/2)}=\frac{2\pi^{1-\epsilon}}{\Gamma(1-\epsilon)}$. Integrating by parts (no surface terms for $\epsilon<0$):

$$\text{trace:}\quad H^{nn}(u)=\mathcal{N}_\epsilon\int\partial_n\phi(r)\,\partial_n\phi(r+u)=-\mathcal{N}_\epsilon\int\phi(r)\nabla^2\phi(r+u)=-\mathcal{N}_\epsilon S_d\,\phi(u)=-\mathcal{N}_\epsilon S_d\frac{|u|^{2\epsilon}}{2\epsilon}, \tag{3.10}$$

$$\text{divergence:}\quad \partial_{u_m}H^{mm'}(u)=-\mathcal{N}_\epsilon\int\nabla^2\phi(r)\,\partial_{m'}\phi(r+u)=-\mathcal{N}_\epsilon S_d\,\frac{u^{m'}}{|u|^d}. \tag{3.11}$$

By rotation covariance and scaling $H^{mm'}=|u|^{2\epsilon}[a\,\delta^{mm'}+b\,\hat u^m\hat u^{m'}]$. Its trace is $(da+b)|u|^{2\epsilon}$ and its divergence is $(2\epsilon a+b)\,u^{m'}|u|^{2\epsilon-2}$ (using $\partial_m(u^mu^{m'}|u|^{2\epsilon-2})=(d-1+2\epsilon)u^{m'}|u|^{2\epsilon-2}=u^{m'}|u|^{2\epsilon-2}$). So $da+b=-\frac{\mathcal{N}_\epsilon S_d}{2\epsilon}$, $2\epsilon a+b=-\mathcal{N}_\epsilon S_d$, i.e. $a=-\frac{\mathcal{N}_\epsilon S_d}{4\epsilon}$, $b=-\frac{\mathcal{N}_\epsilon S_d}{2}$. With $\frac{\mathcal{N}_\epsilon S_d}{2}=\pi\Gamma(1-\epsilon)\big(\frac{\mu^2e^{\gamma_E}}{4}\big)^\epsilon$:

$$\boxed{H^{mm'}(u)=-\pi\,\Gamma(1-\epsilon)\Big(\frac{\mu^2u^2e^{\gamma_E}}{4}\Big)^\epsilon\Big[\frac{\delta^{mm'}}{2\epsilon}+\hat u^m\hat u^{m'}\Big]} \tag{3.12}$$

Expanding, with $\Gamma(1-\epsilon)\big(\frac{\mu^2u^2e^{\gamma_E}}{4}\big)^\epsilon = 1+\epsilon L(u)+O(\epsilon^2)$,

$$L(u)\equiv\log\frac{\mu^2u^2e^{2\gamma_E}}{4},\qquad H^{mm'}(u)=-\frac{\pi}{2}\,\delta^{mm'}\Big[\frac{1}{\epsilon}+L(u)\Big]-\pi\,\hat u^m\hat u^{m'}+O(\epsilon). \tag{3.13}$$

Comparison with my z integral. My identity in (0.15) is exactly this formula with $u\to y'-y$. But $u=(y'-y)/\xi$, and (3.12) gives $H(\xi u)=\xi^{2\epsilon}H(u)$, i.e. $L(\xi u)=L(u)+\log\xi^2$. This is problem (A).

Step 7. Contraction with T, exact in ϵ, N(73)–N(75)

The indices k,k' of C are 2D; $\delta_{mm'}$ is traced in d dimensions; $d_\perp=d=2-2\epsilon$. Write $\Gamma_u\equiv-\pi\Gamma(1-\epsilon)(\mu^2u^2e^{\gamma_E}/4)^\epsilon$.

$$d_\perp\xi(1-\xi)\,\delta_{km}\delta_{k'm'}H^{mm'}C_{kk'}=\Gamma_u\,(2-2\epsilon)\,\xi(1-\xi)\Big[\frac{C_{ii}}{2\epsilon}+\hat u_i\hat u_jC_{ij}\Big]$$

$$= \Gamma_u\,\xi(1-\xi)\Big[\Big(\frac{1}{\epsilon}-1\Big)C_{ii} + 2(1-\epsilon)\hat{u}_i\hat{u}_j C_{ij}\Big], \tag{3.14}$$

$$2\big(\delta_{mk'}\delta_{km'} - \delta_{km}\delta_{k'm'}\big)H^{mm'}C_{kk'} = 2\big(H^{k'k} - H^{kk'}\big)C_{kk'} = 0 \qquad (H \text{ symmetric}), \tag{3.15}$$

$$c(\xi)\,\delta_{mm'}\delta_{kk'}H^{mm'}C_{kk'} = c(\xi)\,\Gamma_u\Big[\frac{d}{2\epsilon}+1\Big]C_{ii} = \Gamma_u\,\frac{c(\xi)}{\epsilon}\,C_{ii}. \tag{3.16}$$

Adding, with $\mathcal{T}(u;\xi) \equiv \hat{u}_i\hat{u}_j C_{ij}(\xi u) - \frac{1}{2}C_{ii}(\xi u)$ and $\hat{P} = \xi(1-\xi)+c$:

$$I(\xi) = -\pi\Gamma(1-\epsilon)\int d^2u\, e^{-ik\cdot u}\Big(\frac{\mu^2u^2e^{\gamma_E}}{4}\Big)^{\epsilon}$$
$$\times\Big\{\Big[\frac{\hat{P}(\xi)}{\epsilon} - \epsilon\,\xi(1-\xi)\Big]C_{ii}(\xi u) + 2(1-\epsilon)\xi(1-\xi)\,\mathcal{T}(u;\xi)\Big\} \tag{3.17}$$

$$= -\pi\int d^2u\, e^{-ik\cdot u}\Big\{\hat{P}(\xi)\Big[\frac{1}{\epsilon}+L(u)\Big]C_{ii}(\xi u) + 2\xi(1-\xi)\,\mathcal{T}(u;\xi)\Big\} + O(\epsilon). \tag{3.18}$$

The $-\xi(1-\xi)C_{ii}$ from $d_\perp/(2\epsilon)$ became $-\epsilon\xi(1-\xi)C_{ii}$ after combining with the $\hat{u}\hat{u}$ term, so the pole coefficient is exactly $\hat{P}$ and no $O(\epsilon)/\epsilon$ remnant survives. (This is the same "$\frac{d_\perp}{2}L_\epsilon = L_\epsilon - 1$" bookkeeping as in my note.)

In my variables, using (3.8):

$$I(\xi) = -\frac{\pi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\hat{P}(\xi)\Big[\frac{2}{\epsilon_{\rm me}} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4\xi^2}\Big]$$
$$\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)$$
$$+\frac{\pi(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - 2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\times\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \tag{3.19}$$

This is N(50). Compared with my (0.15): the second line is identical; in the first line only the ξ^2 differs.

Step 8. The two endpoints, N(77), N(78)

$\xi\to 1$. $(1-\xi)\hat{P}\to 1$, $C(\xi u)\to C(u)$, $(1-\xi)\mathcal{T}\to 0$. Exactly in ϵ:

$$\mathcal{J} \equiv \lim_{\xi\to1}(1-\xi)I(\xi) = -\pi\int d^2u\, e^{-ik\cdot u}\Big[\frac{1}{\epsilon}+L(u)\Big]C_{ii}(u)$$
$$= -\pi\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\Big[\frac{2}{\epsilon_{\rm me}} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}\Big]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)$$
$$= \int_{x,z,w,w',x'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)\,(x-w)\cdot(x'-w')}{(w-z)^2(x-w)^2(w'-z)^2(x'-w')^2}$$
$$\times\big[U^{ce'}(w') - U^{ce'}(x')\big]\big[U^{ce}(w) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{3.20}$$

the last form with the z integral in d dimensions (= N(20); at $\xi=1$, $y=w$, $u=w'-w$, and there is no ξ^2 issue). Its pole $-\frac{\pi}{\epsilon}\mathcal{L}(k)$ is the large-$|z|$ region, the same as the collinear region $|r|\to\infty$.

$\xi\to 0$. $\xi\hat{P}\to 1$, $C(\xi u)\to C(0)$, $\xi\mathcal{T}\to 0$. With $\int d^2u\,e^{-ik\cdot u} = (2\pi)^2\delta^{(2)}(k) = 0$ for $k\neq 0$ and $\int d^2u\,e^{-ik\cdot u}L(u) = -\frac{4\pi}{k^2}$ (the exact version is $\int d^2u\,e^{-ik\cdot u}(u^2)^\epsilon = \frac{4^{1+\epsilon}\pi\Gamma(1+\epsilon)}{\Gamma(-\epsilon)(k^2)^{1+\epsilon}}$):

$$\mathcal{B} \equiv \lim_{\xi\to0}\xi I(\xi) = -\pi\Big(-\frac{4\pi}{k^2}\Big)C_{ii}(0) = \frac{4\pi^2}{k^2}\int_Y A(Y)\cdot A(Y)$$
$$= \frac{4\pi^2}{k^2}\int_{x,x',z}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\big[U^{ce'}(z) - U^{ce'}(x')\big]\big[U^{ce}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)$$
$$= \int_{x,z,w,w',x'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)\,(x-z)\cdot(x'-z)}{(w-z)^2(x-z)^2(w'-z)^2(x'-z)^2}$$
$$\times\big[U^{ce'}(z) - U^{ce'}(x')\big]\big[U^{ce}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{3.21}$$

using $\int_{w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} = \frac{4\pi^2}{k^2}$. $\mathcal{B}$ has no collinear pole.

Step 9. Endpoint subtraction and the ξ integrals, N(24)–N(27)

Define $R(\xi) \equiv I(\xi) - \frac{\mathcal{J}}{1-\xi} - \frac{\mathcal{B}}{\xi}$, integrable on $[0,1]$. Exactly,

$$\int_{\xi_\vee}^{\xi_\Lambda} d\xi\, I(\xi) = \mathcal{J}\log\frac{1-\xi_\vee}{1-\xi_\Lambda} + \mathcal{B}\log\frac{\xi_\Lambda}{\xi_\vee} + \int_{\xi_\vee}^{\xi_\Lambda} d\xi\, R(\xi), \tag{3.22}$$

and with $1-\xi_\Lambda = \frac{\Lambda}{k^+ + \Lambda}$, $\frac{\xi_\Lambda}{\xi_\vee} = \frac{\vee}{k^+ + \Lambda}$:

$$\int_{\xi_\vee}^{\xi_\Lambda} d\xi\, I(\xi) = \log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B} + \int_0^1 d\xi\, R(\xi) + O\Big(\frac{\Lambda}{k^+}, \frac{k^+}{\vee}\Big). \tag{3.23}$$

(My (0.5) used $\frac{\xi}{1-\xi}\mathcal{J}$ and $\frac{1-\xi}{\xi}\mathcal{B}$; the difference $-(\mathcal{J}+\mathcal{B})$ integrates to my boundary terms $\frac{\bar C}{k^+}(\xi_\vee - \xi_\Lambda)(\mathcal{J}+\mathcal{B}) = \bar C[\frac{1}{\vee} - \frac{1}{k^+ + \Lambda}](\mathcal{J}+\mathcal{B})$, which therefore disappear here.)

Step 10. The remainder and the complete coordinate-space result, N(79)–N(82)

Insert (3.18) and the endpoint forms. The $C(0)$ term enters only through $L(u)$ because its $1/\epsilon$ multiplies $\delta^{(2)}(k)$:

$$R(\xi) = -\pi \int d^2u\, e^{-ik\cdot u}\Big\{\Big[\frac{1}{\epsilon} + L(u)\Big]\Delta(\xi,u) + 2\xi(1-\xi)\mathcal{T}(u;\xi)\Big\}, \qquad \Delta(\xi,u) = \hat P(\xi)C_{ii}(\xi u) - \frac{C_{ii}(u)}{1-\xi} - \frac{C_{ii}(0)}{\xi}. \tag{3.24}$$

Δ is bounded at both ends: at $\xi\to 1$ the $\frac{1}{1-\xi}$ cancels; at $\xi\to 0$, $C_{ii}(\xi u) - C_{ii}(0) = O(\xi^2)$ (C_{ii} is even). Collecting Steps 9 and 10:

$$\boxed{\frac{d^3N}{d^3k} = -\frac{\pi\bar C}{k^+}\int d^2u\, e^{-ik\cdot u}\Big\{\Big[\frac{1}{\epsilon} + L(u)\Big]\mathcal{C}(u) + \bar{\mathcal{T}}(u)\Big\}} \tag{3.25}$$

$$\mathcal{C}(u) = \log\frac{k^+}{\Lambda}\,C_{ii}(u) + \log\frac{\vee}{k^+}\,C_{ii}(0) + \int_0^1 d\xi\,\Delta(\xi,u), \qquad \bar{\mathcal{T}}(u) = \int_0^1 d\xi\, 2\xi(1-\xi)\Big[\hat u_i\hat u_j C_{ij}(\xi u) - \tfrac12 C_{ii}(\xi u)\Big]. \tag{3.26}$$

This is N(81)–N(82), valid up to $O(\Lambda/k^+, k^+/\vee)$.

Step 11. All the poles, and the $\overline{\rm MS}$ subtraction

The coefficient of $1/\epsilon$ in (3.25): $\log\frac{k^+}{\Lambda}C_{ii}(u) \to \mathcal{L}(k)$; $C_{ii}(0)\to\delta^{(2)}(k)\to 0$; and

$$\int d^2u\, e^{-ik\cdot u}\int_0^1 d\xi\,\Delta = \int_0^1 d\xi\Big[\hat P\frac{\mathcal{L}(k/\xi)}{\xi^2} - \frac{\mathcal{L}(k)}{1-\xi}\Big] = \int_0^1\frac{d\xi}{\xi^2}\,\hat P_+(\xi)\mathcal{L}(k/\xi), \qquad \mathcal{L}(k/\xi)/\xi^2 = \int d^2u\, e^{-ik\cdot u}C_{ii}(\xi u). \tag{3.27}$$

So the poles are $-\frac{\pi\bar C}{k^+}\frac{1}{\epsilon}\big[\log\frac{k^+}{\Lambda}\mathcal{L}(k) + \int_0^1\frac{d\xi}{\xi^2}\hat P_+\mathcal{L}(k/\xi)\big]$, and the finite parts follow from $\frac{1}{\epsilon} + L(u)\to L(u)$. This gives the note's result N(85):

$$\boxed{\frac{d^3N}{d^3k} = \frac{\bar C}{k^+}\Big\{\log\frac{k^+}{\Lambda}\Big[\mathcal{J}_{\rm fin}(k) - \frac{\pi}{\epsilon}\mathcal{L}(k)\Big] + \log\frac{\vee}{k^+}\mathcal{B}(k) - \frac{\pi}{\epsilon}\int_0^1\frac{d\xi}{\xi^2}\hat P_+(\xi)\mathcal{L}(k/\xi) + F(k;\mu)\Big\} + O\Big(\frac{\Lambda}{k^+}, \frac{k^+}{\vee}\Big)} \tag{3.28}$$

with

$$\mathcal{J}_{\rm fin} = -\pi\int d^2u\, e^{-ik\cdot u}L(u)C_{ii}(u) = -\pi\,\mathcal{G}(1), \qquad \mathcal{B} = -\pi\,\mathcal{G}(0^+), \tag{3.29}$$

$$F(k;\mu) = -\pi\int_0^1 d\xi\Big[\hat P(\xi)\mathcal{G}(\xi) - \frac{\mathcal{G}(1)}{1-\xi} - \frac{\mathcal{G}(0^+)}{\xi}\Big] + \int d^2u\, e^{-ik\cdot u}\big(-\pi\bar{\mathcal{T}}(u)\big), \tag{3.30}$$

$$\begin{aligned}\mathcal{G}(\xi) &\equiv \int d^2u\, e^{-ik\cdot u}L(u)C_{ii}(\xi u) \\ &= \frac{1}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4\,\xi^2}\Big] \\ &\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\end{aligned} \tag{3.31}$$

$$\mathcal{G}(0^+) = -\frac{4\pi}{k^2}\int_{x,x',z}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\big[U^{ce'}(z) - U^{ce'}(x')\big]\big[U^{ce}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x).$$

F is N(34): $F = \int_0^1 d\xi\,[I_{\rm fin} - \mathcal{J}_{\rm fin}/(1-\xi) - \mathcal{B}/\xi]$.

Step 12. Completing P_{gg}: add and subtract $\frac{11}{12}\delta(1-\xi)$, N(38)–N(43)

$P_{gg}(\xi) = 2N_c\big[\hat{P}_+(\xi) + \frac{11}{12}\delta(1-\xi)\big]$ ($n_f = 0$, $\beta_0 = \frac{11N_c}{3}$), the $\frac{11}{12}$ being fixed by $\int_0^1 d\xi\,\xi\hat{P}_+ = -\frac{11}{12}$. Hence

$$-\frac{\pi}{\epsilon}\int_0^1 \frac{d\xi}{\xi^2}\hat{P}_+\mathcal{L}(k/\xi) = -\frac{\pi}{\epsilon}\int_0^1 \frac{d\xi}{\xi^2}\Big[\hat{P}_+(\xi) + \frac{11}{12}\delta(1-\xi)\Big]\mathcal{L}(k/\xi) + \frac{\pi}{\epsilon}\frac{11}{12}\mathcal{L}(k). \tag{3.32}$$

The first term is the full-P_{gg} $\overline{\text{MS}}$ fragmentation counterterm, the second cancels the α_s counterterm ($\frac{\alpha_s}{2\pi}2N_c\frac{11}{12} = \frac{\alpha_s\beta_0}{4\pi}$), and $-\frac{\pi}{\epsilon}\log\frac{k^+}{\Lambda}\mathcal{L}(k)$ is the UV of target evolution.

Step 13. Net-colour IR pole, N(56)–N(63), N(83)–N(84)

With $Q^c \equiv \int_x[\delta^{ce} - U^{ce}(x)]\rho^e(x)$, $A_i^c(Y) \to -Y_iQ^c/Y^2$ at large $|Y|$, so (cut $|Y| < R_{\text{IR}}$) $C_{ij}(v) \to \pi\delta_{ij}Q^cQ^c\log R_{\text{IR}}$, $C_{ii}(v) \to 2\pi Q^cQ^c\log R_{\text{IR}}$, independent of v. Then $\bar{\mathcal{T}}$ gets nothing ($\hat{u}_i\hat{u}_j\delta_{ij} - \frac{1}{2}\delta_{ii} = 0$), and $\mathcal{C}(u)$ gets the constant $2\pi Q^2\log R_{\text{IR}}\big[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} + \int_0^1 d\xi(\hat{P} - \frac{1}{1-\xi} - \frac{1}{\xi})\big]$ with $\int_0^1(\xi(1-\xi) - 2) = -\frac{11}{6}$. Multiplying $L(u)$ and Fourier transforming ($-4\pi/k^2$):

$$\frac{d^3N}{d^3k} \supset \frac{\bar{C}}{k^+}\frac{8\pi^3Q^cQ^c}{k^2}\log R_{\text{IR}}\Big[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} - \frac{11}{6}\Big] \xrightarrow{\log R_{\text{IR}}\to-\frac{1}{2\epsilon}} -\frac{\bar{C}}{k^+}\frac{\pi}{\epsilon}\mathcal{L}_C(k)\Big[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} - \frac{11}{6}\Big], \quad \mathcal{L}_C = \frac{4\pi^2Q^cQ^c}{k^2}. \tag{3.33}$$

Removing it from $\mathcal{J}_{\text{fin}}, \mathcal{B}, F$ defines $\hat{\mathcal{J}}, \hat{\mathcal{B}}, \hat{F}$, and (3.28)+(3.32)+(3.33) is N(86):

$$\begin{aligned}\frac{d^3N}{d^3k} = \frac{\bar{C}}{k^+}\Big\{ &-\frac{\pi}{\epsilon}\int_0^1\frac{d\xi}{\xi^2}\Big[\hat{P}_+ + \tfrac{11}{12}\delta(1-\xi)\Big]\mathcal{L}(k/\xi) + \frac{\pi}{\epsilon}\frac{11}{12}\mathcal{L}(k) - \frac{\pi}{\epsilon}\log\frac{k^+}{\Lambda}\mathcal{L}(k)\\ &- \frac{\pi}{\epsilon}\mathcal{L}_C(k)\Big[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} - \frac{11}{6}\Big] + \log\frac{k^+}{\Lambda}\hat{\mathcal{J}}(k) + \log\frac{\vee}{k^+}\hat{\mathcal{B}}(k) + \hat{F}(k;\mu)\Big\}.\end{aligned} \tag{3.34}$$

4 Final result in my notation ($\frac{1}{\epsilon} \to \frac{2}{\epsilon_{\text{me}}}$) — identical to N(85)/N(86)

$$\frac{d^3N_{\text{NLO}}}{d^3k} = \frac{d^3N_{\text{DGLAP}}}{d^3k} + \frac{d^3N_{\text{RC}}}{d^3k} + \frac{d^3N_{\text{finite}}}{d^3k} + \frac{d^3N_{\text{BFKL}}}{d^3k} + \frac{d^3N_{\text{JIMWLK}}}{d^3k}, \tag{4.1}$$

$$\begin{aligned}\frac{d^3N_{\text{DGLAP}}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\frac{2}{\epsilon_{\text{me}}}\int_0^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\\ &\times\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\end{aligned} \tag{4.2}$$

$$\mathcal{P}_{gg}(\xi) = 2N_c\Big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\Big] + \frac{\beta_0}{2}\,\delta(1-\xi), \qquad \beta_0 = \frac{11N_c}{3},$$

$$\begin{aligned}\frac{d^3N_{\text{RC}}}{d^3k} =& +\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\beta_0}{16\pi^2}\frac{1}{k^+}\frac{2}{\epsilon_{\text{me}}}\\ &\times\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\end{aligned} \tag{4.3}$$

$$\begin{aligned}\frac{d^3N_{\text{finite}}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\left\{\frac{\xi\,\mathcal{G}(\xi) - \mathcal{G}(1)}{1-\xi}\right.\\ &\left.+\frac{(1-\xi)\,\mathcal{G}(\xi) - \mathcal{G}(0^+)}{\xi} + \xi(1-\xi)\,\mathcal{G}(\xi)\right\}\\ &+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\\ &\times\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - 2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\ &\times\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x),\end{aligned} \tag{4.4}$$

$$\frac{d^3N_{\text{BFKL}}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{k^+}\int_{x,z,w,w',x'} e^{-ik(w'-w)}\frac{(w-z)\cdot(w'-z)\,(x-z)\cdot(x'-z)}{(w-z)^2(x-z)^2(w'-z)^2(x'-z)^2}$$

$$\times \big[U^{ce'}(z) - U^{ce'}(x')\big]\big[U^{ce}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{4.5}$$

$$\begin{aligned}
\frac{d^3N_{\text{JIMWLK}}}{d^3k} &= + \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,z,w,w',x'} e^{-ik(w'-w)}\frac{(w-z)\cdot(w'-z)\,(x-w)\cdot(x'-w')}{(w-z)^2(x-w)^2(w'-z)^2(x'-w')^2} \\
&\quad\times\big[U^{ce'}(w') - U^{ce'}(x')\big]\big[U^{ce}(w) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x) \\
&= - \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\Big[\frac{2}{\epsilon_{\text{me}}} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}\Big] \\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\big[U^{ce'}(y') - U^{ce'}(x')\big]\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x),
\end{aligned} \tag{4.6}$$

with $\mathcal{G}$ as in (3.31); the ξ^2 in $\mathcal{G}(\xi)$ is the correction of problem (A). Term by term: DGLAP = "full P_{gg}: $\overline{\text{MS}}$ FF counterterm" of N(86); RC = "cancels α_s counterterm"; JIMWLK = $\log\frac{k^+}{\Lambda}[\mathcal{J}_{\text{fin}} - \frac{\pi}{\epsilon}\mathcal{L}]$ (contains the "UV of target evolution" pole); BFKL = $\log\frac{\vee}{k^+}\mathcal{B}$; finite = $F(k;\mu)$. If $Q^c \neq 0$, JIMWLK, BFKL and finite contain the net-colour pole (3.33), which in my convention reads $-\frac{1}{(2\pi)^3}\frac{g^4N_c}{2\pi^2}\frac{1}{k^+k^2}\frac{2}{\epsilon_{\text{me}}}Q^cQ^c\big[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} - \frac{11}{6}\big]$; pulling it out gives N(86) exactly.

After counterterms (N Sec. 7.5, end of 9.6): the $\overline{\text{MS}}$ FF counterterm at μ_F removes the DGLAP pole and leaves $P_{gg}\otimes\log(\mu_F^2u^2e^{2\gamma_E}/4)$ acting on $C_{ii}(\xi u)$; the coupling counterterm at μ_R removes the RC pole and its $\frac{11}{12}$ share of $\log u^2$ at $\xi = 1$ gives $\alpha_s\big(4e^{-2\gamma_E}/|y-y'|^2\big)$ inside the LO integral, N(55).